

Adjoint-Based Aerodynamic Shape Optimization for the Discontinuous Galerkin Approach

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Introduction

Introduction

- ▶ Most commercial CFD software today rely on low-order methods (methods with an order of accuracy of at most 2).
 - **Order of Accuracy:** The rate at which the error of the solution decreases as the mesh is refined.
 - To get a solution with low error, need a very fine mesh (requires a very high computational cost).
- ▶ This has resulted in increased research into **high-order** methods for CFD analysis.
 - One such high-order method is the Discontinuous Galerkin (DG) Method, which is the focus of this work.

Introduction (2)

- ▶ Aerodynamic design optimization is a key application of CFD.
 - A powerful approach to optimize is through using the adjoint based approach.
 - Developed extensively and available in commercial low-order CFD softwares (ex. ANSYS Adjoint Solver)
- ▶ **Research Goal:** Develop an adjoint solver for aerodynamic shape optimization for a high-order Discontinuous Galerkin CFD code.

The Discontinuous Galerkin (DG) Method

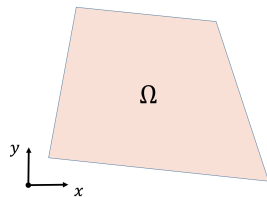
The 2D Scalar Conservation Equation

- ▶ Consider the 2D Scalar Conservation Equation given by

$$\frac{\partial U}{\partial t} + \nabla \cdot \mathbf{F} = 0$$

where $\mathbf{F} = (F, G)$, $F = F(U)$ and $G = G(U)$.

- ▶ The goal is to search for a numerical approximate solution $U^h(x, y, t)$ to the given PDE on a computational domain Ω



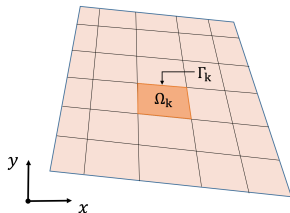
The 2D Scalar Conservation Equation

Solving using the DG Method

- ▶ In the DG Method, discretize the domain Ω into N_e elements.
- ▶ The PDE will then be solved to find the approximate solution (U_k^h) within **each** element Ω_k .
- ▶ The collection of local solutions makes up the total numerical solution

$$U^h(x, y, t) = \bigoplus_{k=1}^{N_e} U_k^h(x, y, t)$$

- ▶ In the DG Method, the solutions $U_k^h(x, y, t)$ are **not** required to be continuous at the element boundaries (Γ_k).



2D Scalar Conservation Equation

Solving using the DG Method (2)

- ▶ To represent the solution on each element, use N_p independent basis functions $w_i(x, y)$

$$U_k^h(x, y, t) = \sum_{i=1}^{N_p} a_i^k(t) w_i(x, y)$$

- Initial Condition: $U(x, y, 0)$ known $\rightarrow$ Find $a_i^k(0) \rightarrow$ Know $U_k^h(x, y, 0)$ for all elements.
- Use procedure outlined to find a_i^k at next time step and repeat.
- ▶ Form the residual

$$\underbrace{\frac{\partial U}{\partial t} + \nabla \cdot \mathbf{F}(U)}_{\text{Residual } R(x, y, t)} = 0 \rightarrow R^h(x, y, t) = \frac{\partial U^h}{\partial t} + \nabla \cdot \mathbf{F}(U^h) \approx 0$$

- ▶ We require that the residual be orthogonal to each basis function on the element of interest

$$\langle R^h, w_j \rangle = \int_{\Omega_k} R^h(x, y, t) w_j(x, y) d\Omega = 0, \quad j = 1, \dots, N_p$$

The 2D Scalar Conservation Equation

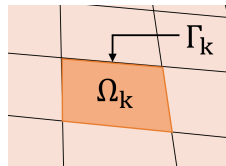
Solving using the DG Method (3)

- ▶ Substituting the expression $U_k^h(x, y, t) = \sum_{i=1}^{N_p} a_i^k(t) w_i(x, y)$ into the residual and simplifying, the **weak form** of the DG scheme is obtained:

$$\sum_{i=1}^{N_p} \left[\frac{da_i^k}{dt} \int_{\Omega_k} w_i w_j d\Omega \right] - \int_{\Omega_k} \nabla w_j \cdot \mathbf{F}(U^h) d\Omega + \int_{\Gamma_k} w_j (\mathbf{F}(U^h) \cdot \hat{n})^* d\Gamma = 0, \quad j = 1, \dots, N_p$$

- ▶ This is a set of N_p equations for N_p unknowns $(\frac{da_i^k}{dt})$ for the solution on element Ω_k .
- ▶ Solve the given system in each element for the $\frac{da_i^k}{dt}$ terms. The value for the coefficients at the next time step can then be found using

$$a_i^{k,n+1} = a_i^{k,n} + \Delta t \cdot \frac{da_i^k}{dt}$$



Basis Functions

Bezier Basis

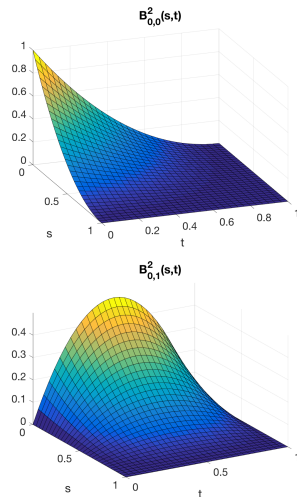
- ▶ Used to represent the numerical solution in all cases.
- ▶ One Dimension: For an order p , there are $p + 1$ basis functions of the form

$$B_i^p(s) = \binom{p}{i} s^i (1-s)^{p-i}, \quad s \in [0, 1], \quad i = 0, \dots, p$$

- ▶ Two Dimensions: For an order p , there are $(p + 1)^2$ basis functions of the form

$$B_{i,j}^p(s, t) = B_i^p(s) B_j^p(t), \quad s, t \in [0, 1]$$

- ▶ Order of Accuracy = $P + 1$
- ▶ $P = 1 \rightarrow$ Equivalent to low order method
- ▶ $P > 1 \rightarrow$ High order method



Example of $P=2$ Bezier Basis Functions

The Two Dimensional Euler Equations

- ▶ The 2D Euler Equations are the flow equations that have been solved for all cases. They are given by

$$\frac{\partial \vec{U}}{\partial t} + \frac{\partial \vec{F}}{\partial x} + \frac{\partial \vec{G}}{\partial y} = 0$$

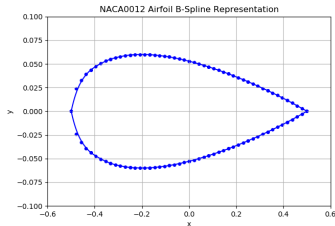
where

$$\vec{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho e \end{bmatrix}, \quad \vec{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho ue + pu \end{bmatrix}, \quad \vec{G} = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho ve + pv \end{bmatrix}$$

Aerodynamic Shape Optimization

Overview

- ▶ Aerodynamic shape optimization requires:
 - A cost function I to be minimized (example: drag coefficient)
 - A set of design variables (α_i)
- ▶ The geometry for all test cases was specified using a B-spline and its control points (these control points were used as the design variables α_i for the optimization)
- ▶ **Goal:** Find the positions to place the design variables α_i to minimize I



Overview (2)

- ▶ To minimize a given cost function I , the gradient with respect to the N_d design variables is needed

$$\nabla_{\alpha} I = \left[\frac{\partial I}{\partial \alpha_1}, \dots, \frac{\partial I}{\partial \alpha_{N_d}} \right]$$

- ▶ With the gradient, move in the negative gradient direction to compute the updated position of the design variables

$$\alpha^{n+1} = \alpha^n - \lambda \nabla_{\alpha} I$$

where $\alpha = [\alpha_1, \dots, \alpha_{N_d}]$

- ▶ Simple approach to compute gradient:

$$\frac{\partial I}{\partial \alpha_i} \approx \frac{I(\alpha_i + \Delta \alpha_i) - I(\alpha_i)}{\Delta \alpha_i}$$

Major Disadvantage: Requires N_d flow solves (each flow solve can be very expensive)

The Adjoint Approach

Provides an approach to compute gradients of the cost function I with a computational cost as expensive as **one** flow solve.

Cost Function

$$I = I(\mathbf{w}, \boldsymbol{\alpha})$$

where $\mathbf{w}$ consist of the N_f flow variables and $\boldsymbol{\alpha}$ consists of the N_d the design variables.

Flow Constraint Equations

Flow variables $\mathbf{w}$ must satisfy the flow constraint equations (ex. Euler Equations) in their discretized form

$$\mathbf{R} = \mathbf{R}(\mathbf{w}, \boldsymbol{\alpha}) = 0$$

where $\mathbf{R}$ is a vector-valued function consisting of N_f components.

The Adjoint Approach (2)

The gradient of the cost function can now be computed by solving the **Adjoint equation** and then substituting that solution into the **Sensitivity equation**.

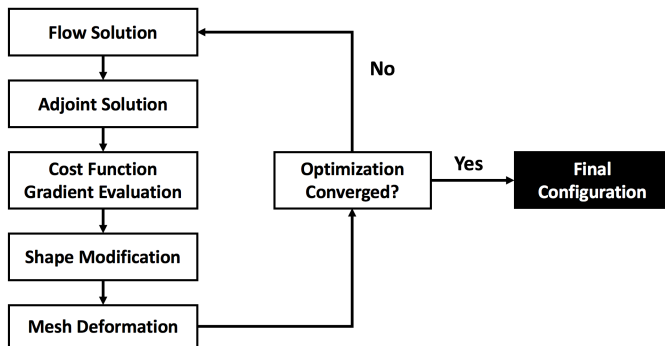
Adjoint Equation

$$\left[\frac{\partial \mathbf{R}}{\partial \mathbf{w}} \right]^T \boldsymbol{\psi} = \left[\frac{\partial I}{\partial \mathbf{w}} \right]$$

Sensitivity Equation

$$\nabla_{\alpha} I = \left[\frac{\partial I}{\partial \alpha} \right]^T - \boldsymbol{\psi}^T \left[\frac{\partial \mathbf{R}}{\partial \alpha} \right]$$

Optimization Procedure



Case: Inverse Design - NACA0012 to NACA4412

Inverse Design - NACA0012 to NACA4412

Goal: From an initial airfoil, try to get to the target airfoil shape.

Cost Function

$$I = \frac{1}{2} \int_S (P - P_{target})^2 dS$$

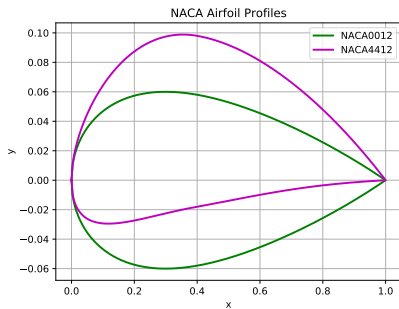
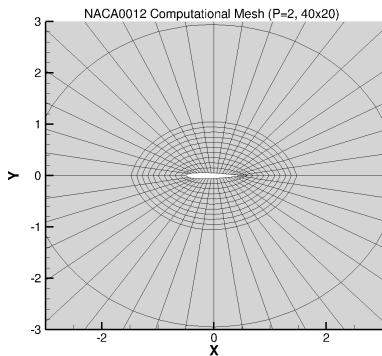
where S is the design airfoil surface and P_{target} is the desired surface pressure distribution.

Case Parameters

- ▶ **Initial Profile** = NACA0012 Airfoil
- ▶ **Target Profile** = NACA4412 Airfoil
- ▶ Mach number = 0.1, $\alpha = 0^\circ$
- ▶ $P = 2$, Mesh Dimension = 40 x 20. 78 Design Variables with y direction degree of freedom (Leading, Trailing Edge Fixed)

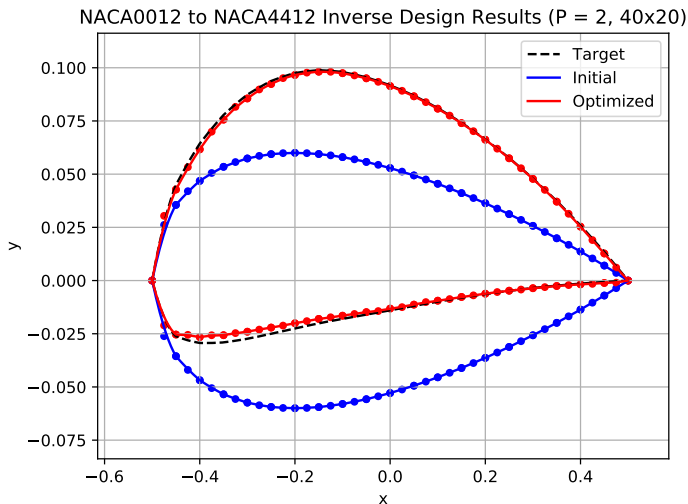
Inverse Design - NACA0012 to NACA4412

Results



Inverse Design - NACA0012 to NACA4412

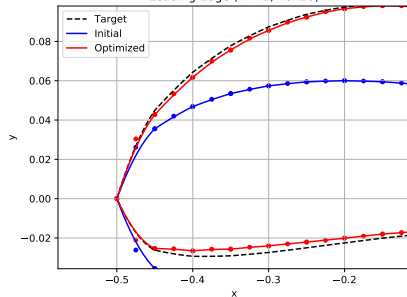
Results (2)



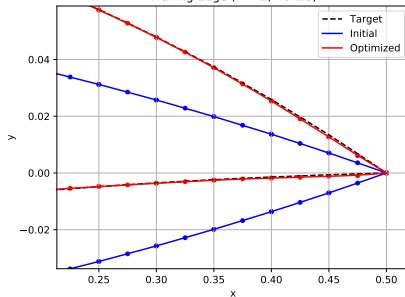
Inverse Design - NACA0012 to NACA4412

Results (3)

NACA0012 to NACA4412 Inverse Design Results
Leading Edge ($P = 2$, 40×20)

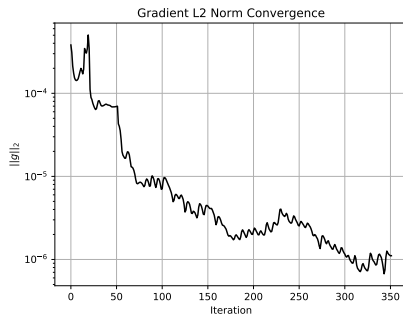
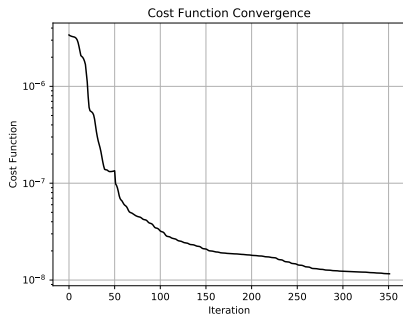


NACA0012 to NACA4412 Inverse Design Results
Trailing Edge ($P = 2$, 40×20)



Inverse Design - NACA0012 to NACA4412

Results (4)



Case: Target Lift Coefficient - NACA0012

Target Lift Coefficient - NACA0012 Airfoil

Goal: From an initial airfoil, try to achieve the target lift coefficient.

Cost Function

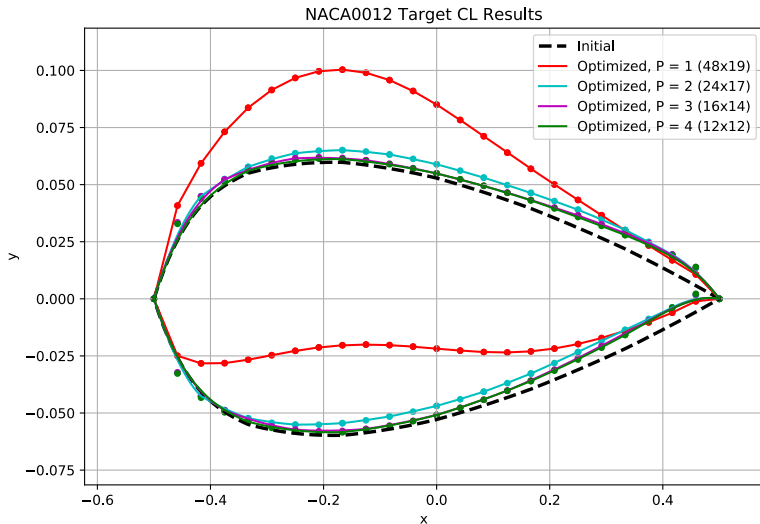
$$I = (CL - CL_{target})^2$$

Case Parameters

- ▶ Mach number = 0.1, $\alpha = 0^\circ$
- ▶ $CL_{target} = 0.24$
- ▶ Design Variables = 46 Control Points with y direction degree of freedom (Leading, Trailing Edge Fixed)

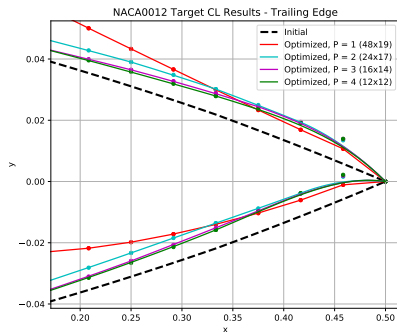
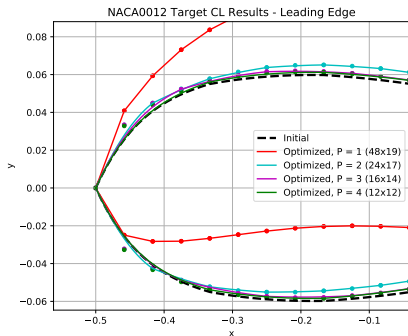
Target Lift Coefficient - NACA0012 Airfoil

Results



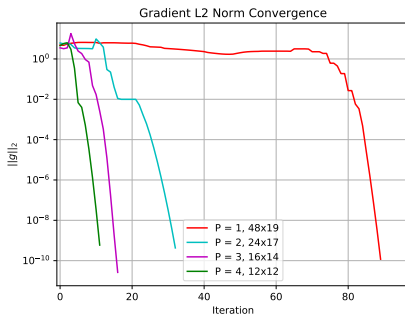
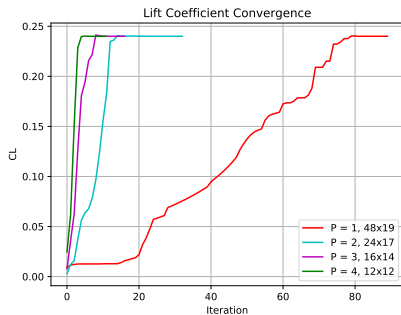
Target Lift Coefficient - NACA0012 Airfoil

Results (2)



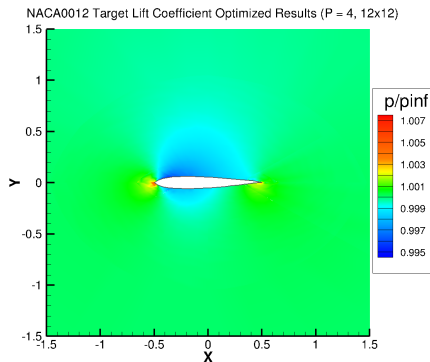
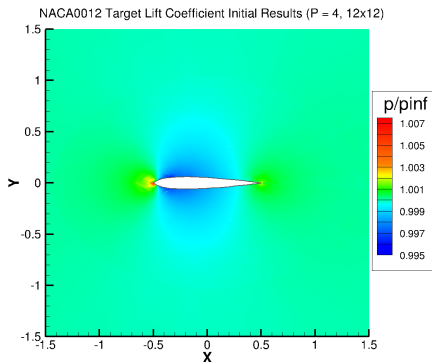
Target Lift Coefficient - NACA0012 Airfoil

Results (3)



Target Lift Coefficient - NACA0012 Airfoil

Results (4)



Conclusion

- ▶ The adjoint approach has proven to be an effective method to optimize geometries with the high-order Discontinuous Galerkin Method.
- ▶ The coarser meshes than can be used for high-order approaches can allow for quicker convergence to optimal solutions.
- ▶ Future Work
 - Extension to solve problems in 3D.
 - Implementation of NURBS basis functions (Isogeometric Analysis).

Questions?

Motivation

Simple Example

- Consider two vectors $\vec{v}_1$ and $\vec{v}_2$ that form a basis for a vector space.
- Interested in finding the best approximation ($\vec{y}_h$) for $\vec{y}$ in the vector space.

$$\vec{y}_h = a\vec{v}_1 + b\vec{v}_2$$

- Galerkin Approach to find a and b : Require the error $\vec{e} = \vec{y} - \vec{y}_h$ to be orthogonal to all basis vectors

$$\langle \vec{e}, \vec{v}_1 \rangle = 0$$

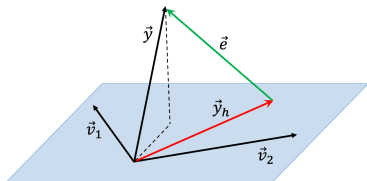
$$\langle \vec{e}, \vec{v}_2 \rangle = 0$$

- The resulting system of linear equations is given by

$$\langle \vec{y}, \vec{v}_1 \rangle - a\langle \vec{v}_1, \vec{v}_1 \rangle - b\langle \vec{v}_1, \vec{v}_2 \rangle = 0$$

$$\langle \vec{y}, \vec{v}_2 \rangle - a\langle \vec{v}_1, \vec{v}_2 \rangle - b\langle \vec{v}_2, \vec{v}_2 \rangle = 0$$

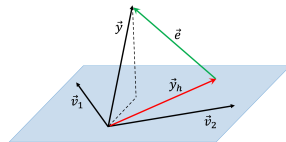
which can be solved for a and b



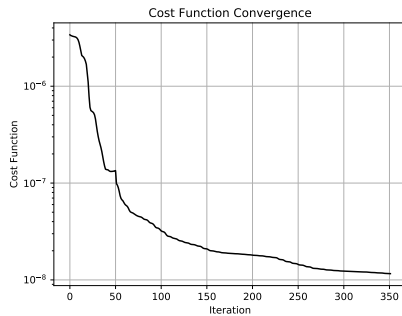
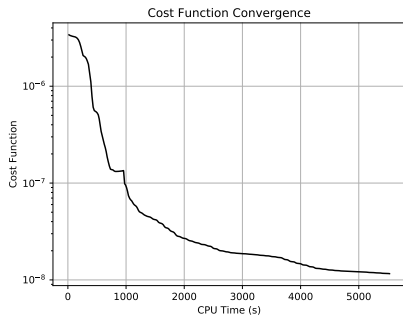
The 2D Scalar Conservation Equation

Solving using the DG Method (Summary)

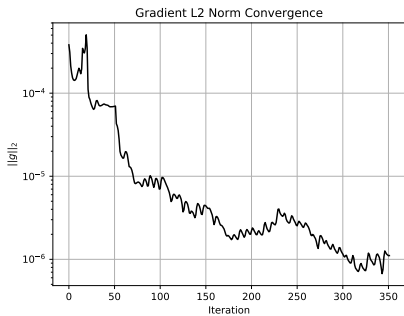
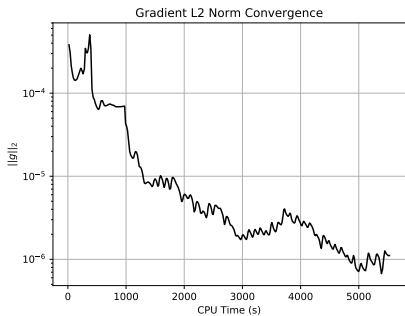
	DG Method (on element Ω_k)	Vector Space Approximation Example
Exact Solution	$U_k(x, y, t)$	$\vec{y}$
Approximate Solution	$U_k^h(x, y, t)$	$\vec{y}_h$
Basis	$w_i(x, y) \quad i = 1, \dots, N_p$	$\vec{v}_1, \vec{v}_2$
Approximate Solution Representation	$U_k^h(x, y, t) = \sum_{i=1}^{N_p} a_i^k(t) w_i(x, y)$	$\vec{y}_h = a \vec{v}_1 + b \vec{v}_2$
Inner Product	$\langle r_1, r_2 \rangle = \int_{\Omega_k} r_1 r_2 d\Omega$	$\langle \vec{r}_1, \vec{r}_2 \rangle = \vec{r}_1 \cdot \vec{r}_2$
Error	$R^h(x, y, t) = \frac{\partial U^h}{\partial t} + \nabla \cdot \mathbf{F}(U^h)$	$\vec{e} = \vec{y} - \vec{y}_h$
Output	$\frac{da_i^k}{dt}, a_i^{k,n} \Rightarrow a_i^{k,n+1}$	a, b



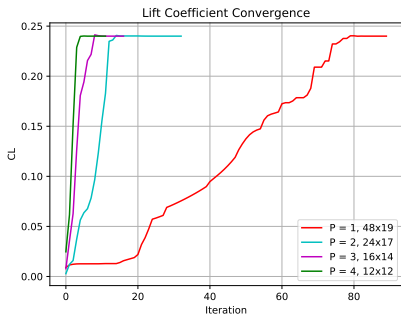
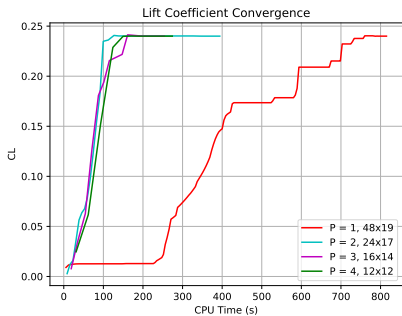
Inverse Design - NACA0012 to NACA4412



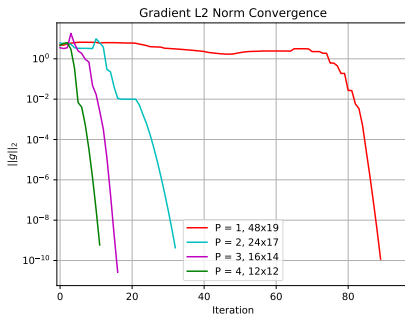
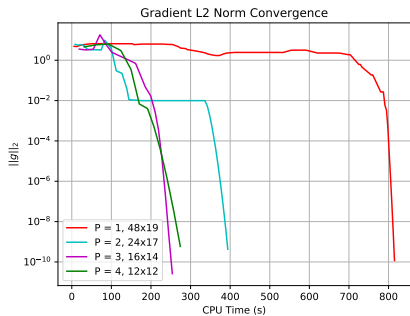
Inverse Design - NACA0012 to NACA4412



Target Lift Coefficient - NACA0012 Airfoil



Target Lift Coefficient - NACA0012 Airfoil



Target Lift Coefficient - NACA0012 Airfoil

